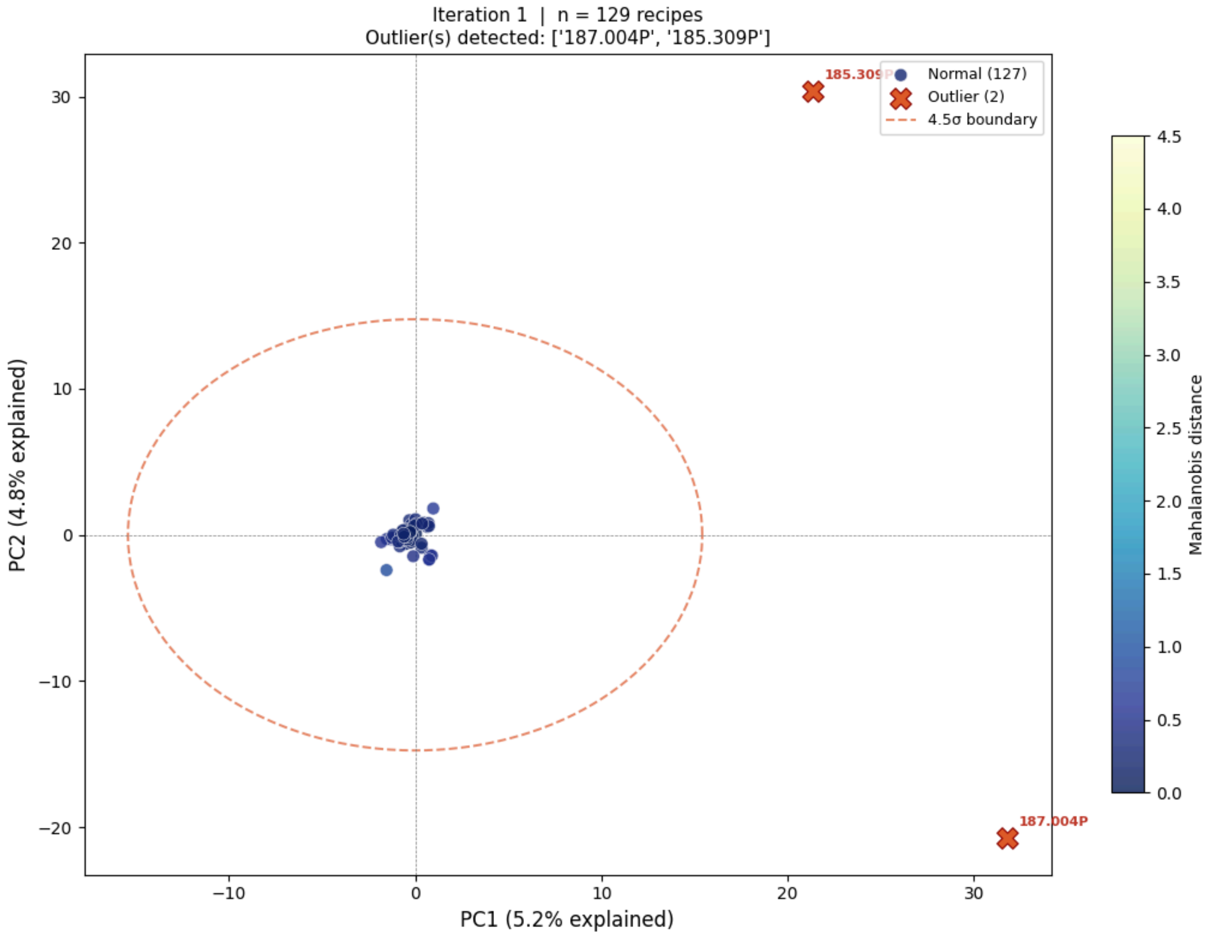
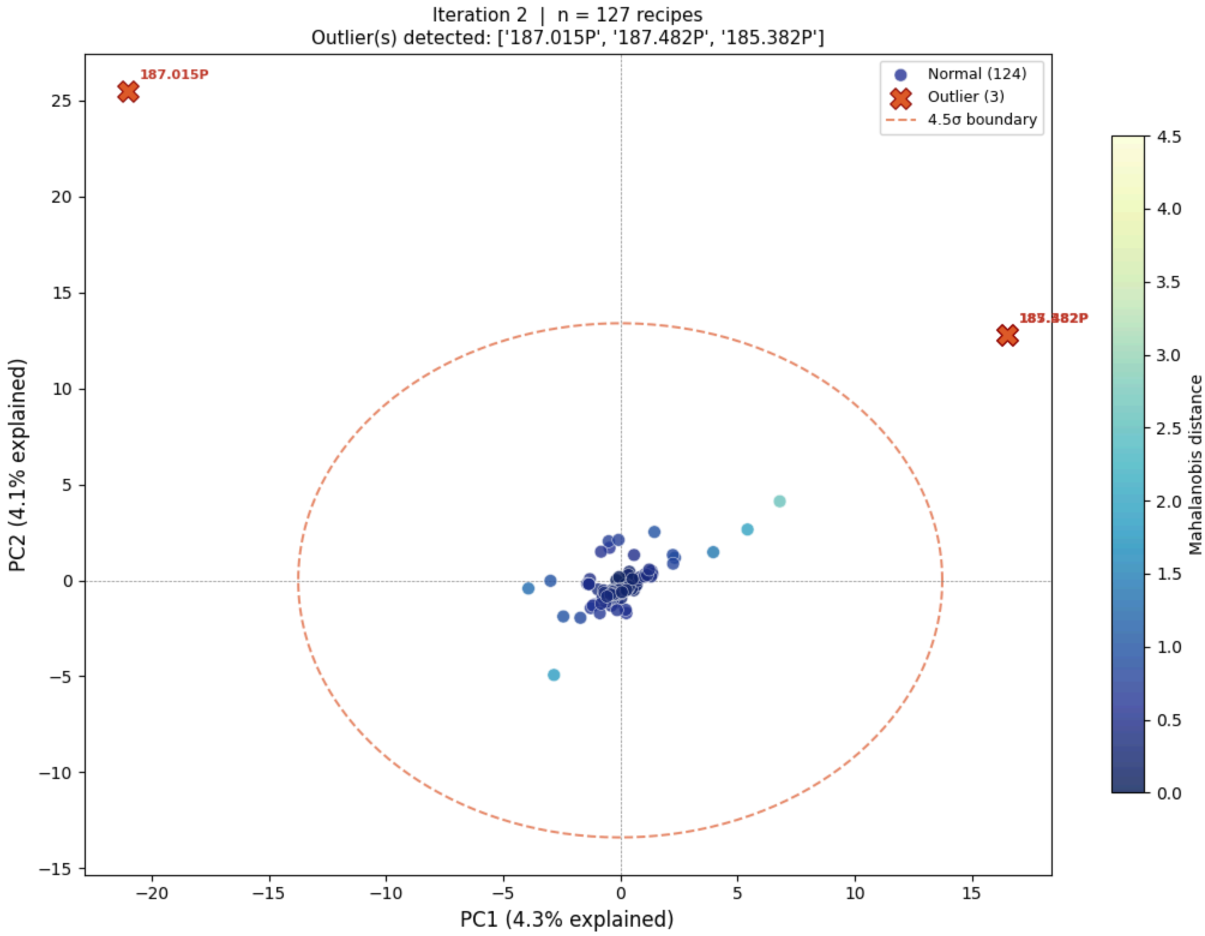
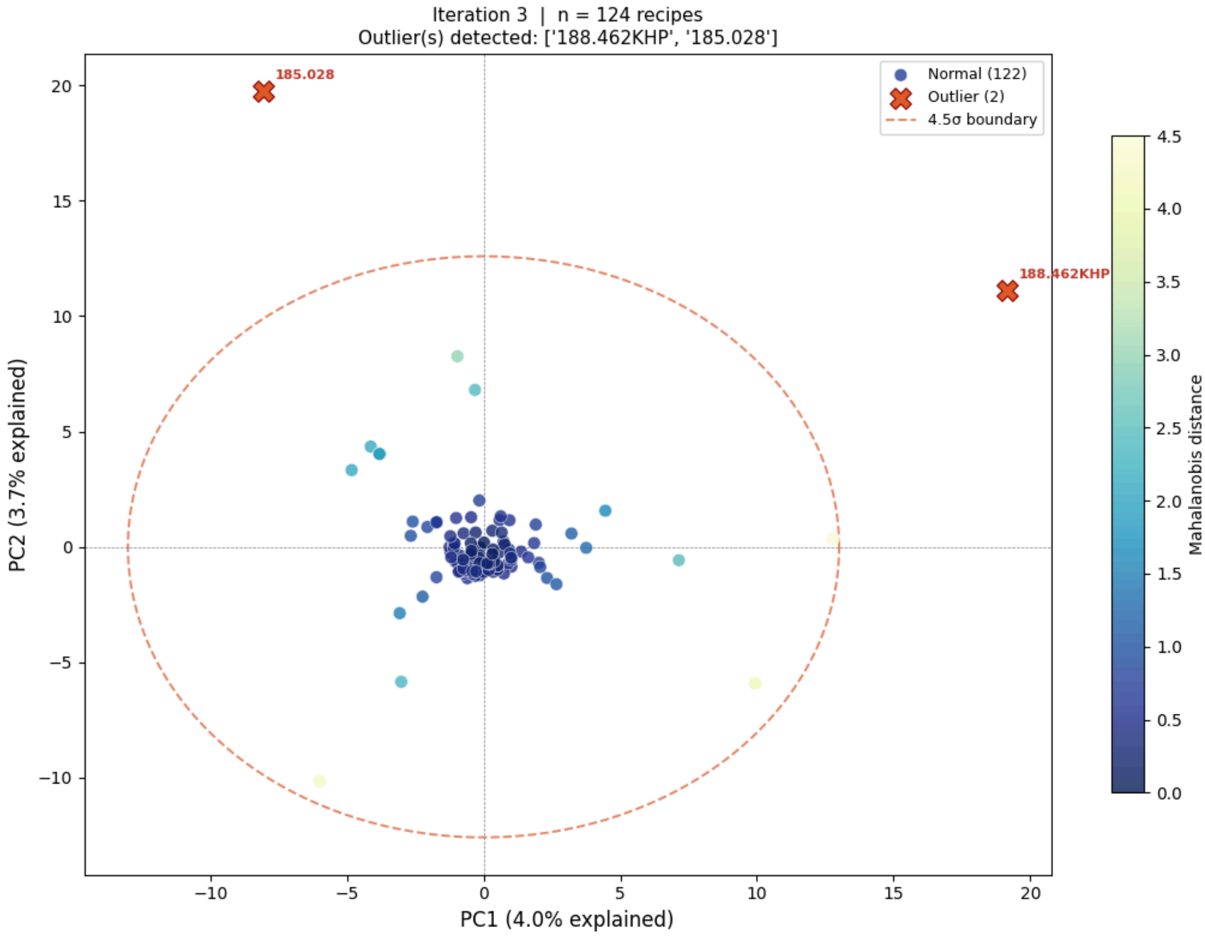


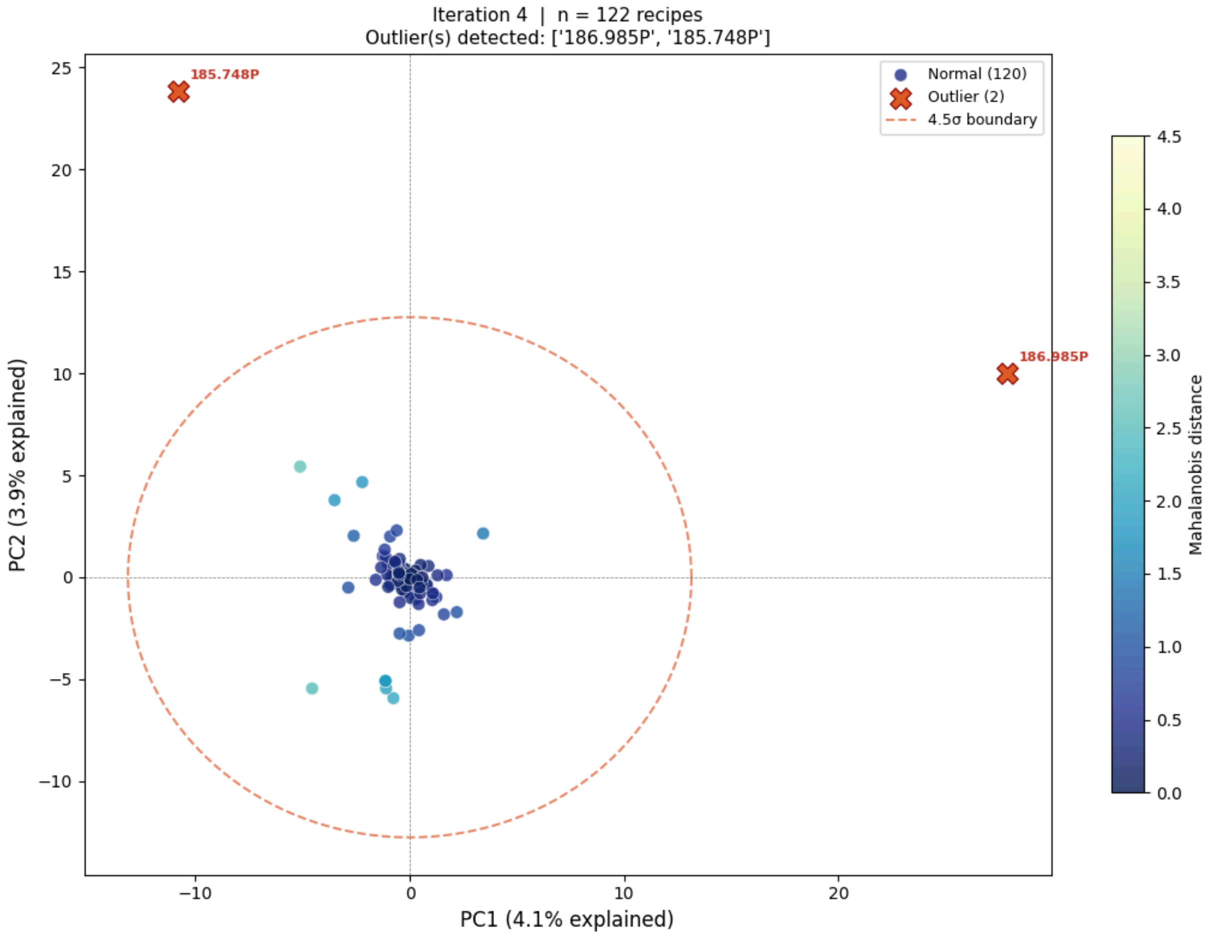

```
Saved: pca_v2_noOT_iter01.png
Iteration 1: removing 2 outlier(s): ['187.004P', '185.309P']
```



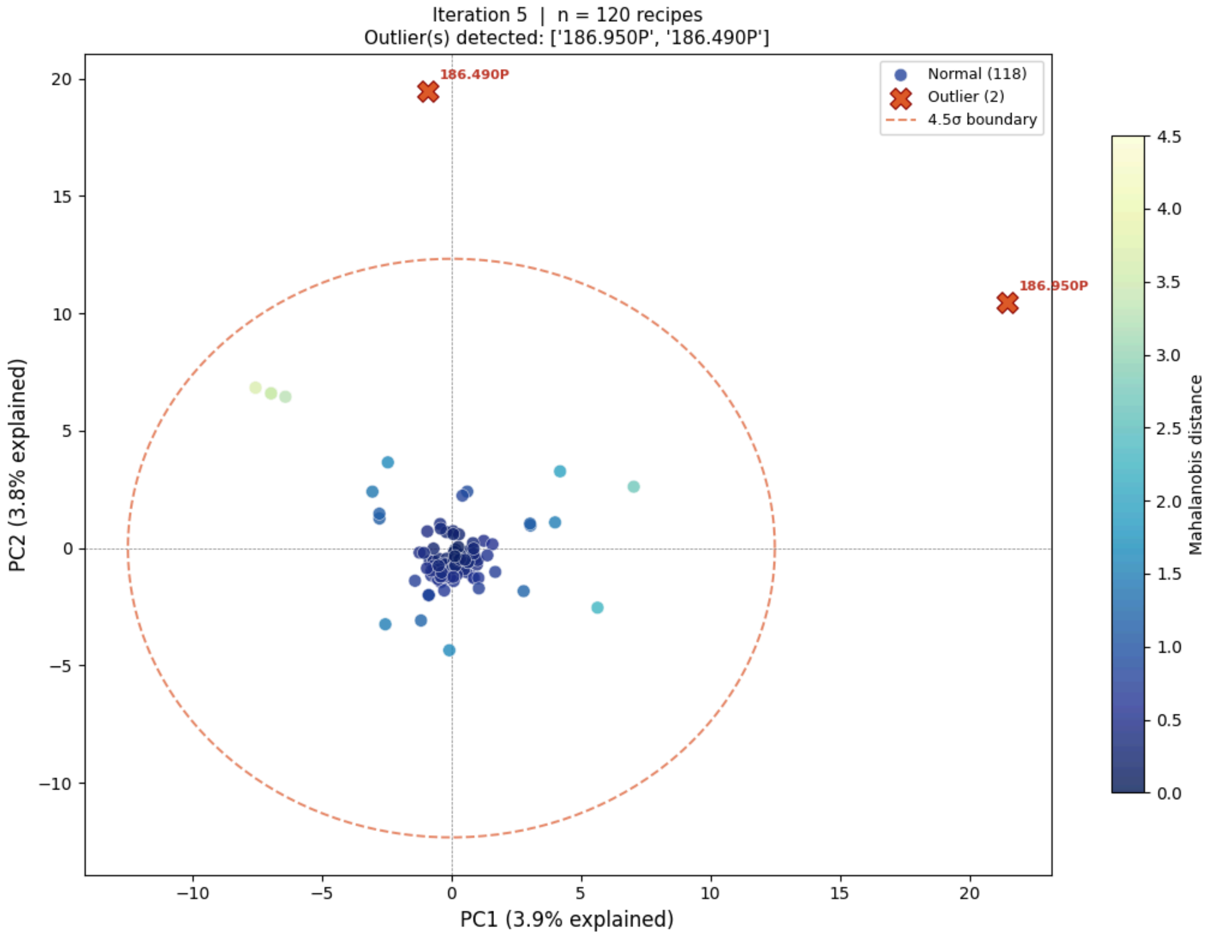

```
Saved: pca_v2_noOT_iter02.png
Iteration 2: removing 3 outlier(s): ['187.015P', '187.482P', '185.382P']
```



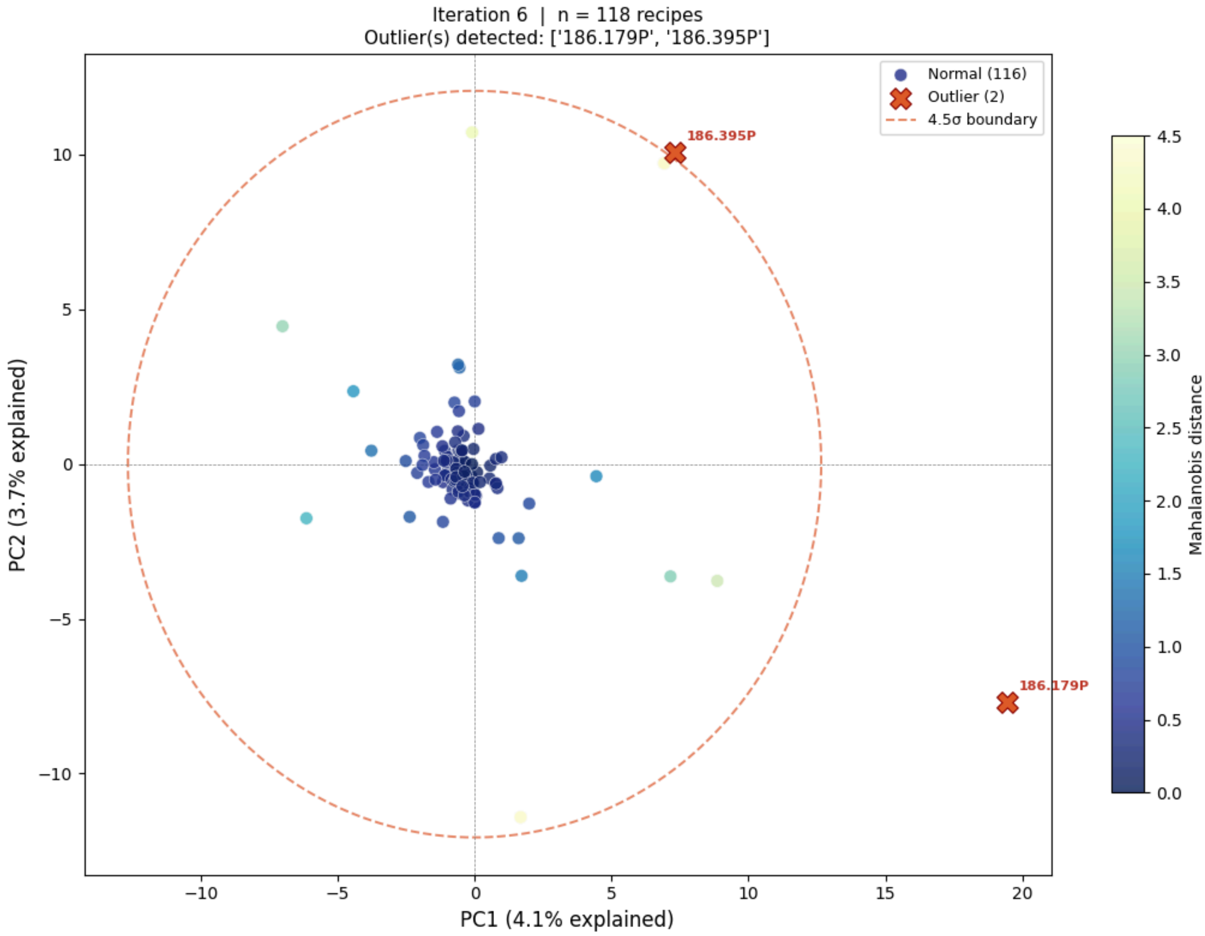

```
Saved: pca_v2_noOT_iter03.png
Iteration 3: removing 2 outlier(s): ['188.462KHP', '185.028']
```



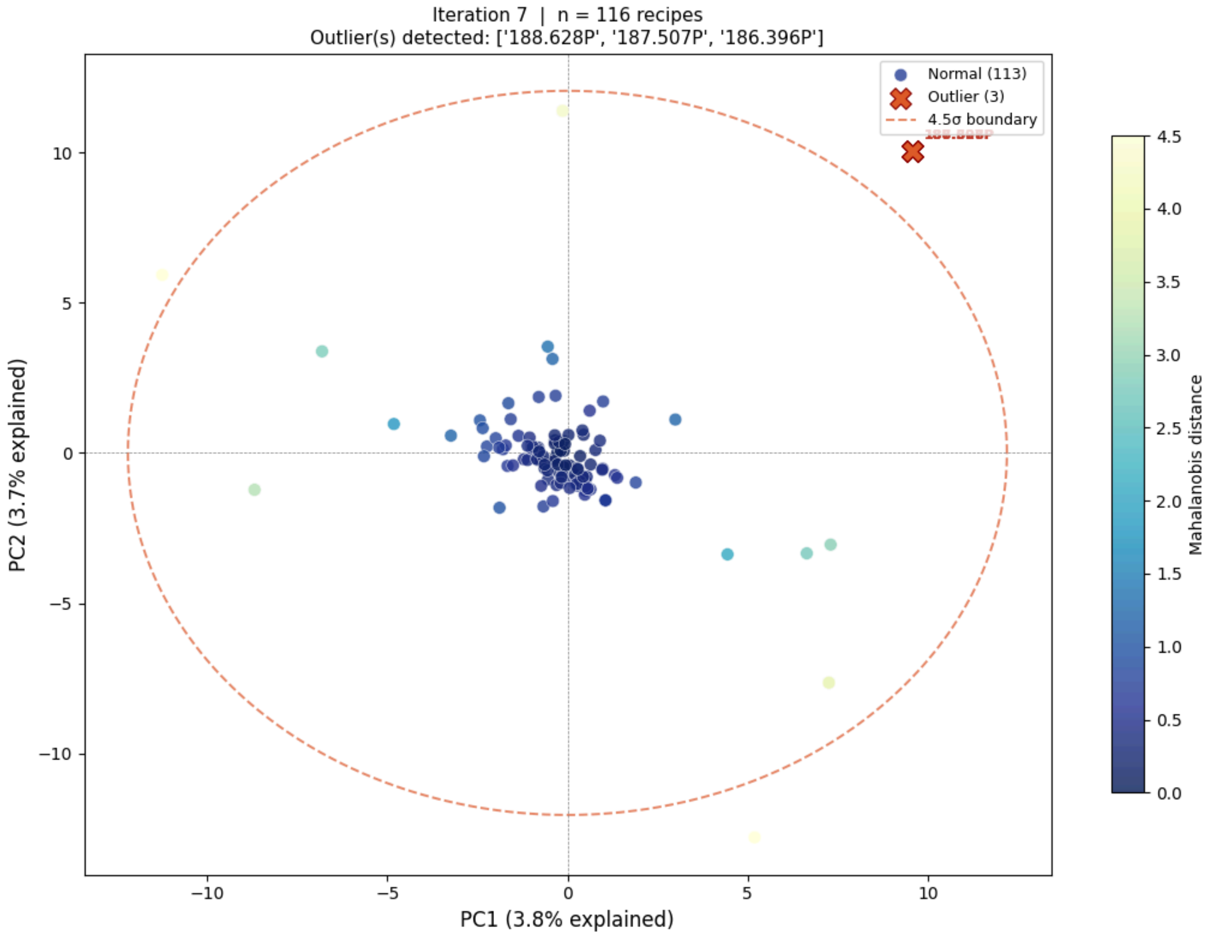

```
Saved: pca_v2_noOT_iter04.png
Iteration 4: removing 2 outlier(s): ['186.985P', '185.748P']
```



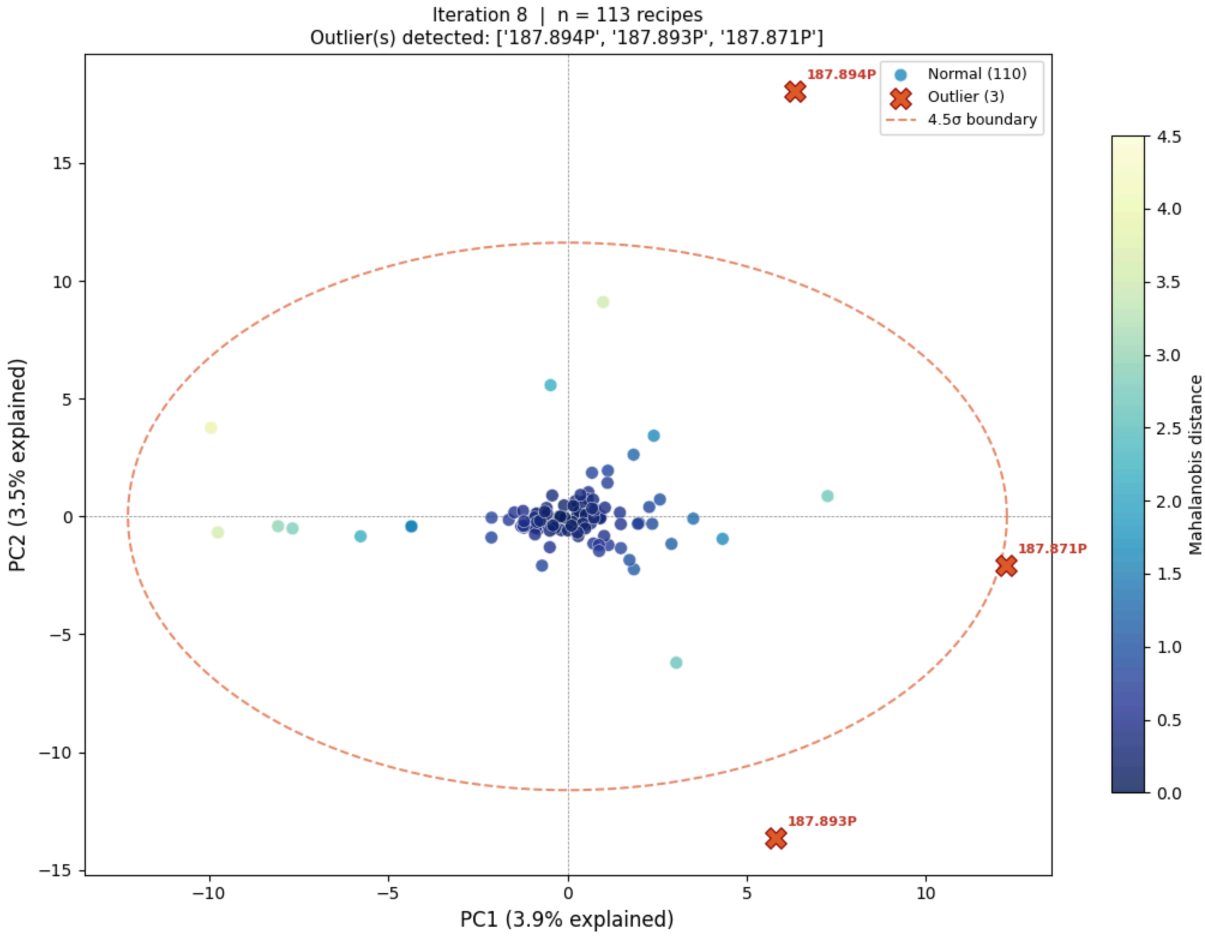

```
Saved: pca_v2_noOT_iter05.png
Iteration 5: removing 2 outlier(s): ['186.950P', '186.490P']
```



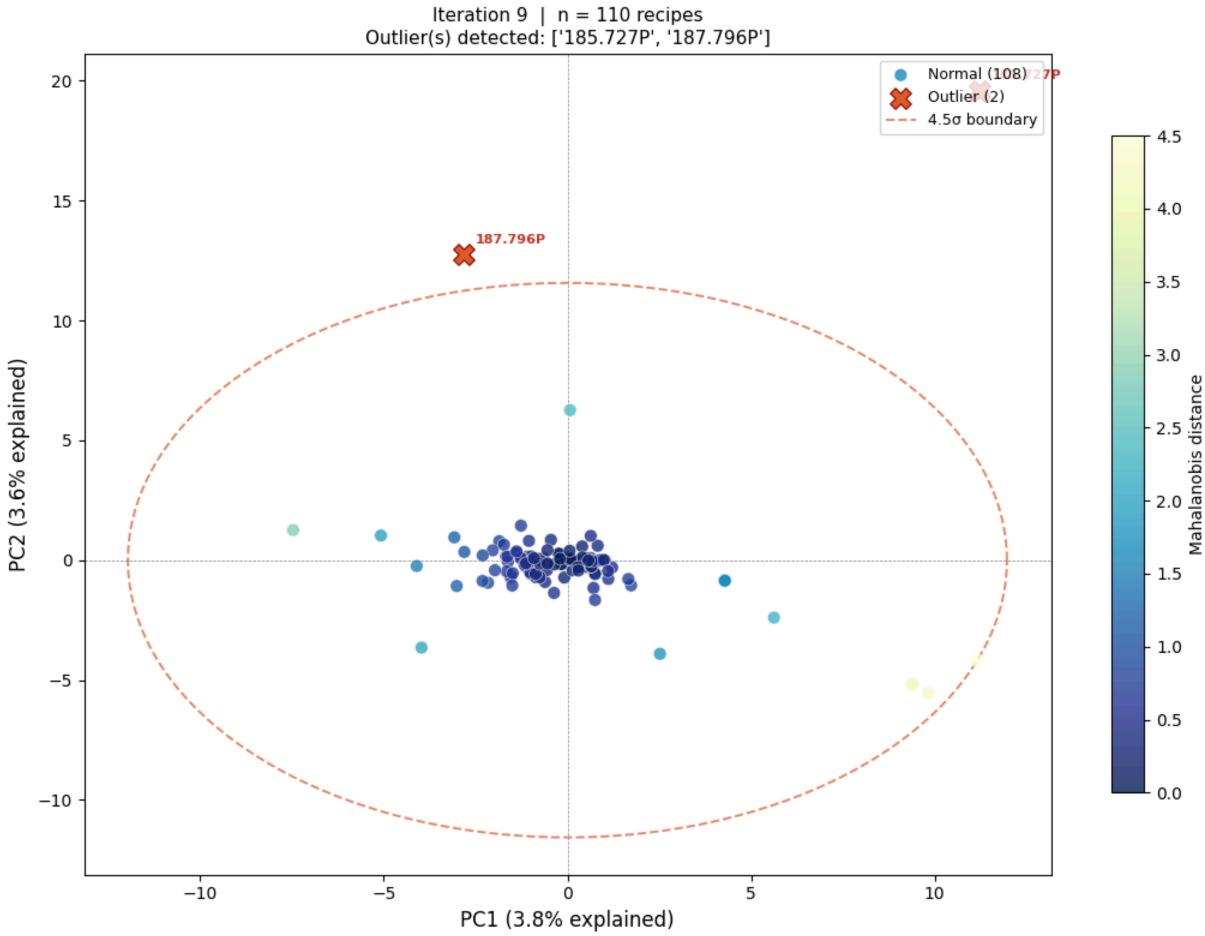

```
Saved: pca_v2_noOT_iter06.png
Iteration 6: removing 2 outlier(s): ['186.179P', '186.395P']
```



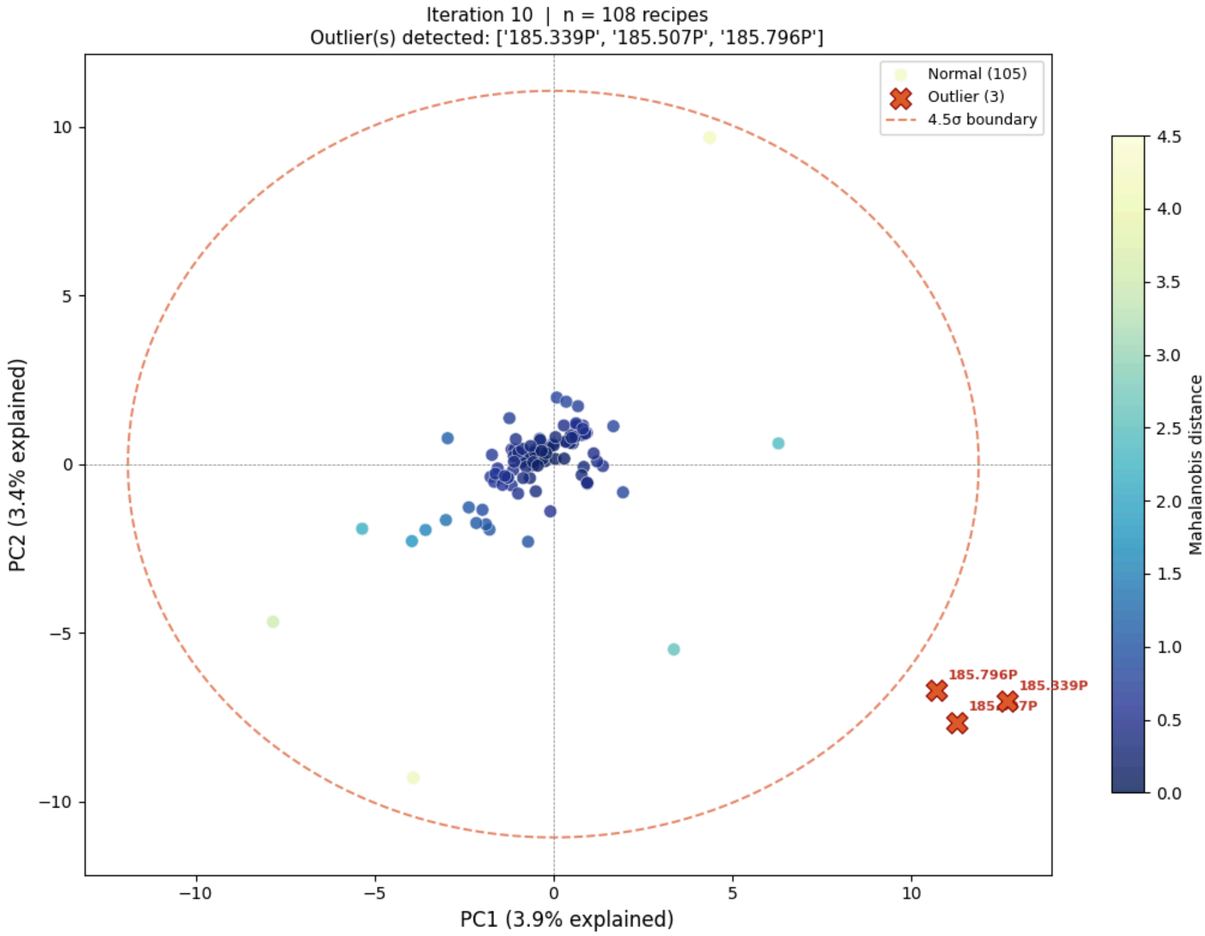

```
Saved: pca_v2_noOT_iter07.png
Iteration 7: removing 3 outlier(s): ['188.628P', '187.507P', '186.396P']
```



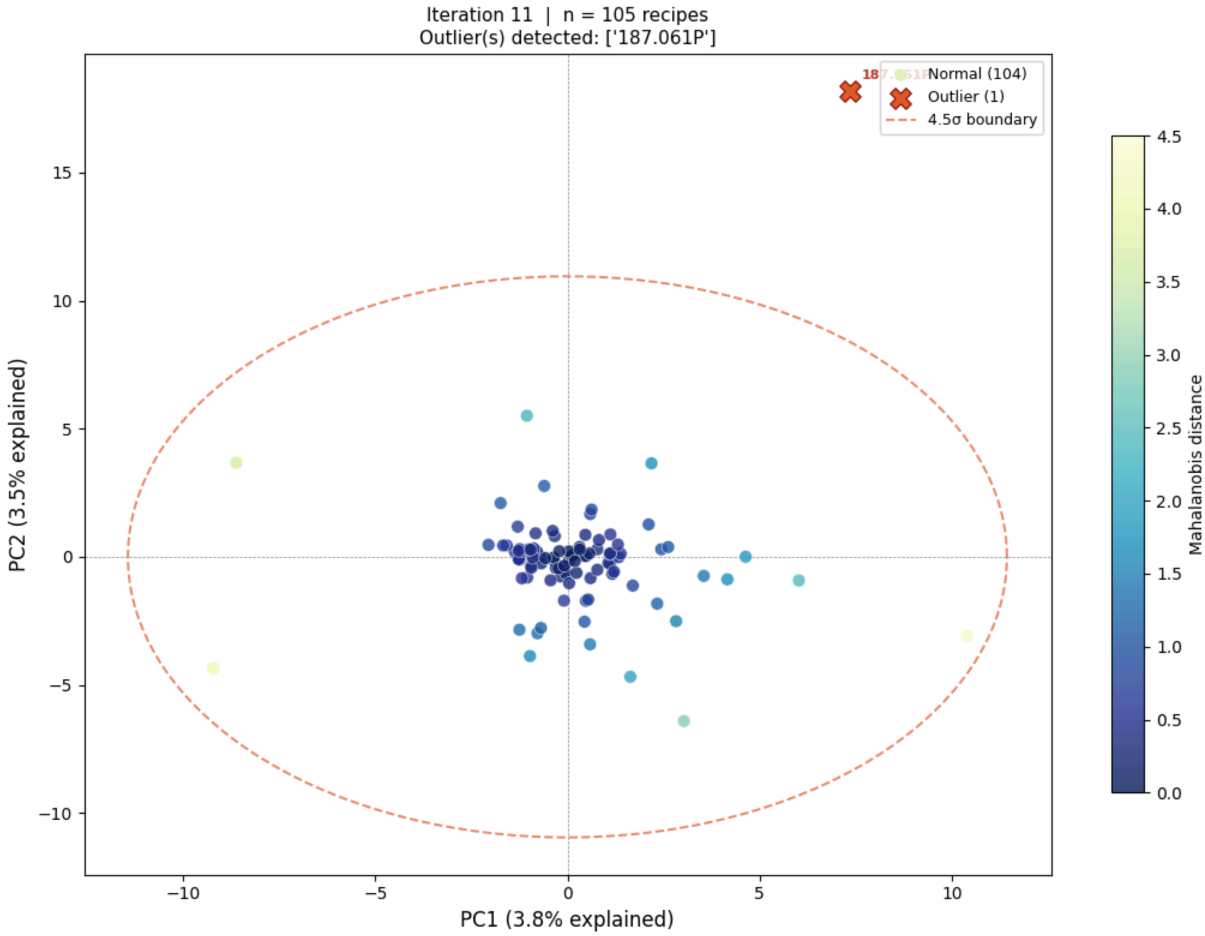

```
Saved: pca_v2_noOT_iter08.png
Iteration 8: removing 3 outlier(s): ['187.894P', '187.893P', '187.871P']
```



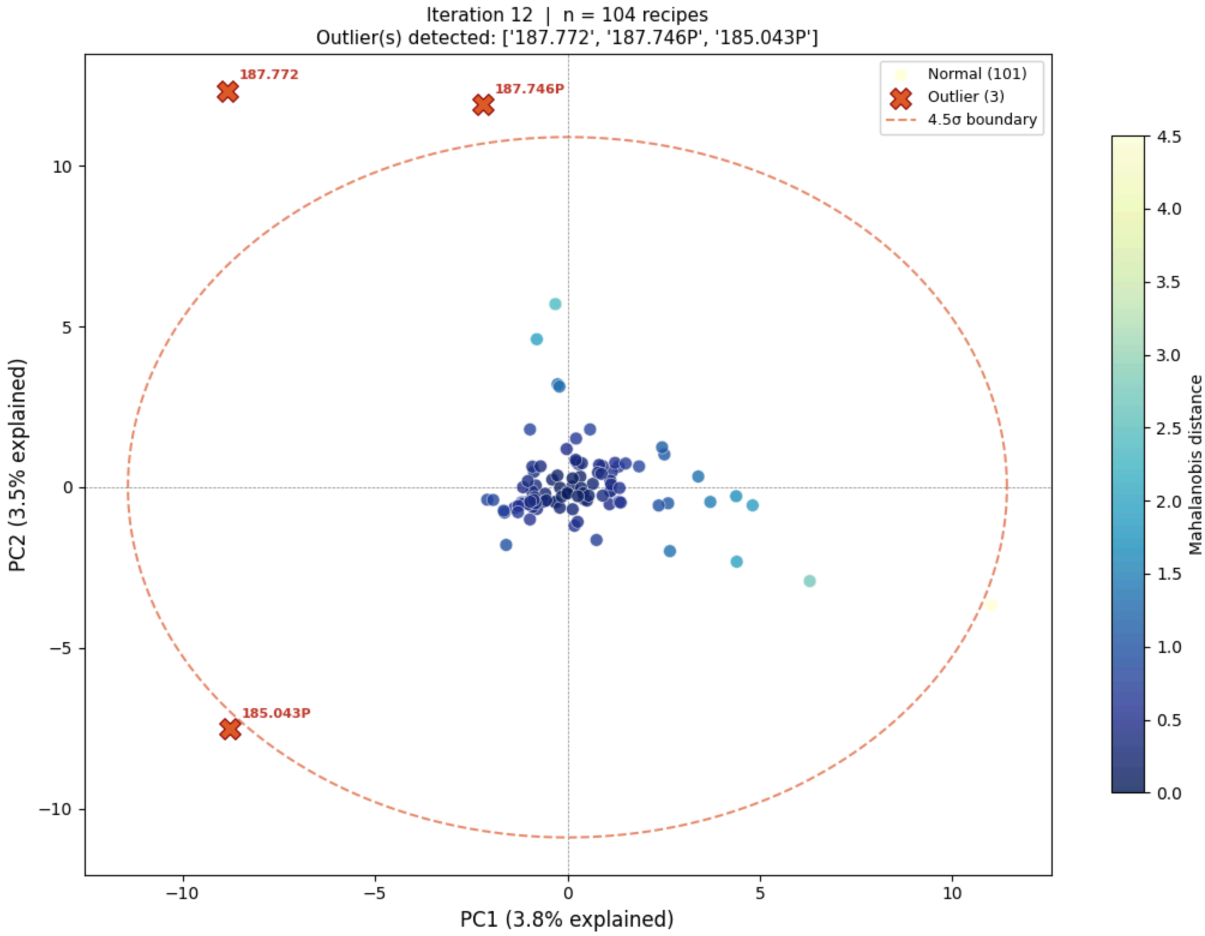

```
Saved: pca_v2_noOT_iter09.png
Iteration 9: removing 2 outlier(s): ['185.727P', '187.796P']
```



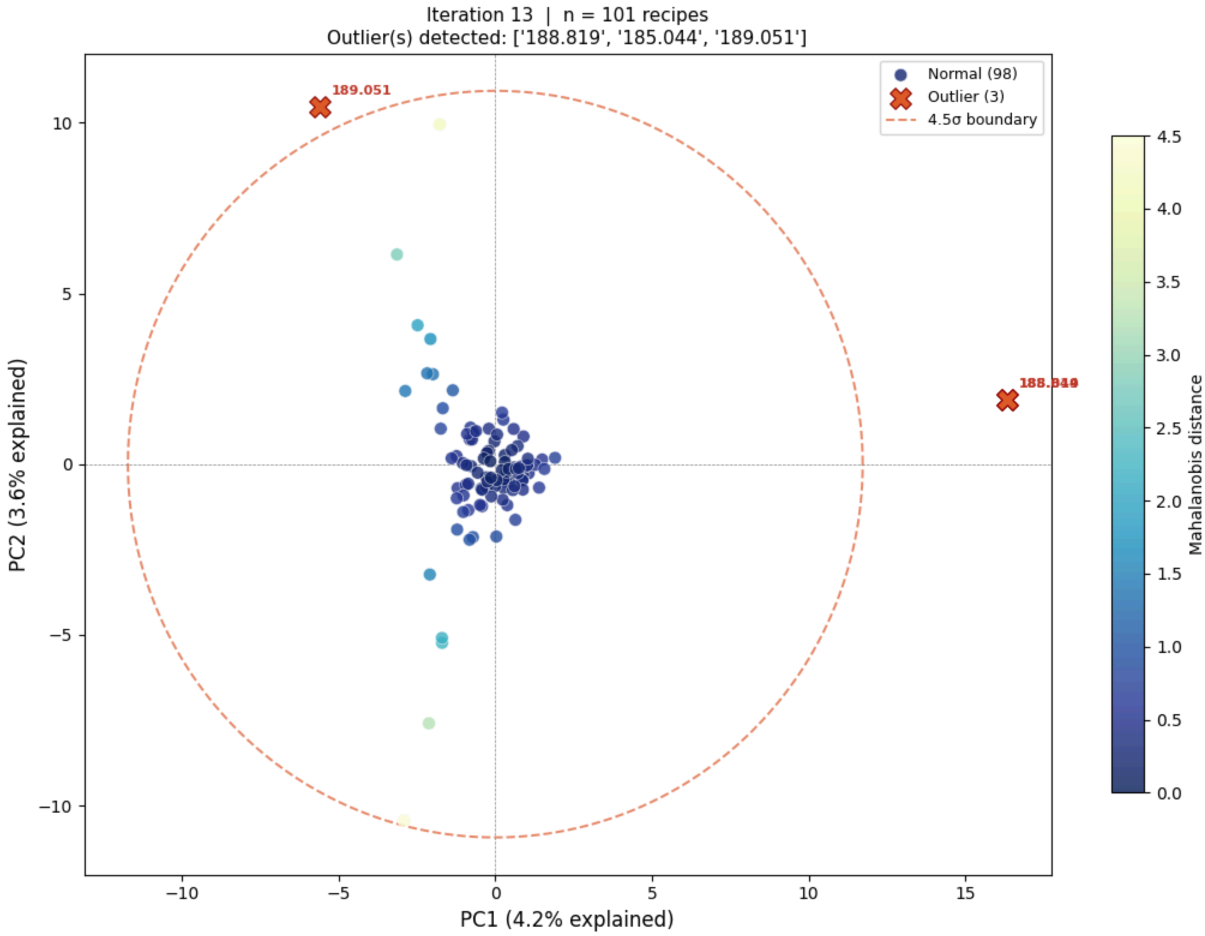

```
Saved: pca_v2_noOT_iter10.png
Iteration 10: removing 3 outlier(s): ['185.339P', '185.507P', '185.796P']
```



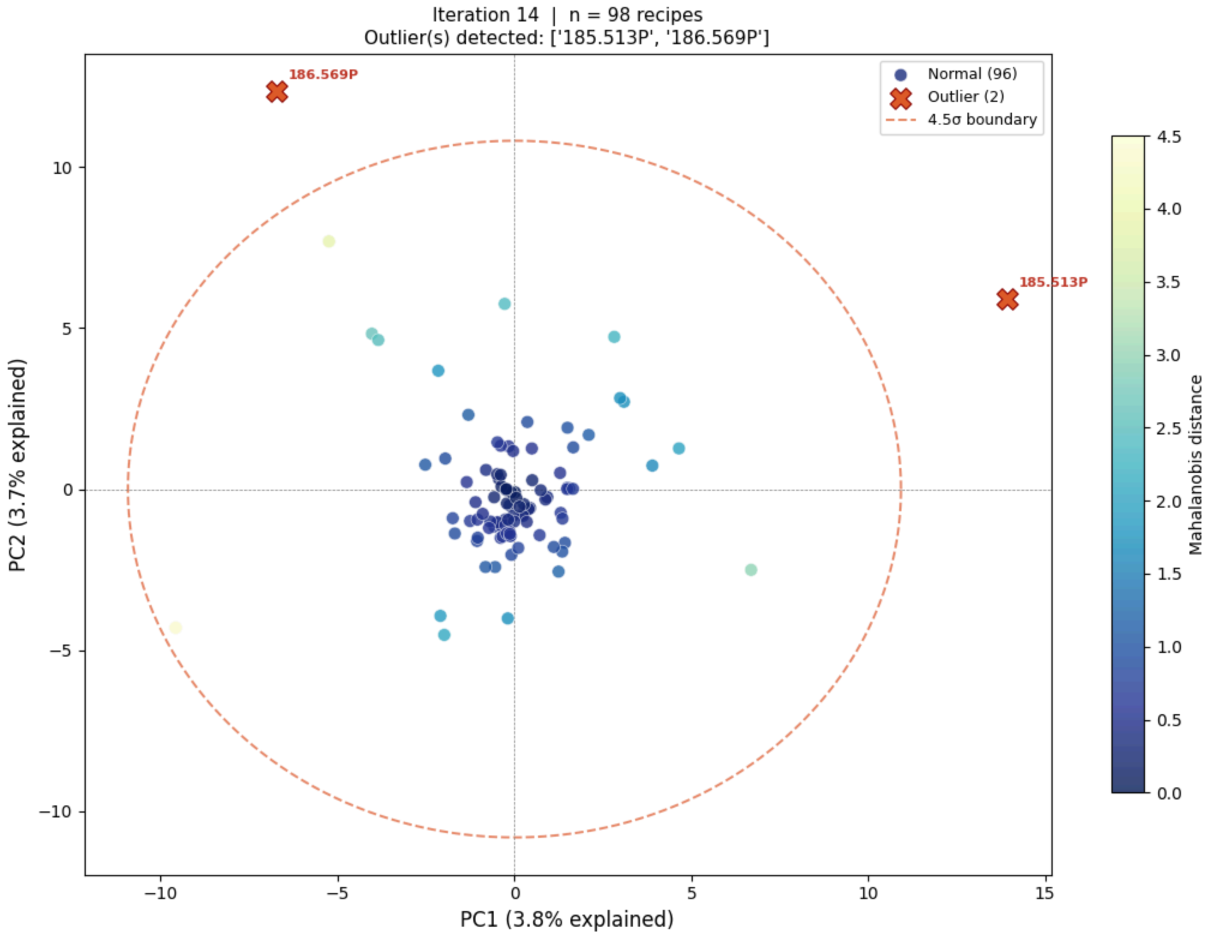

```
Saved: pca_v2_noOT_iter11.png
Iteration 11: removing 1 outlier(s): ['187.061P']
```



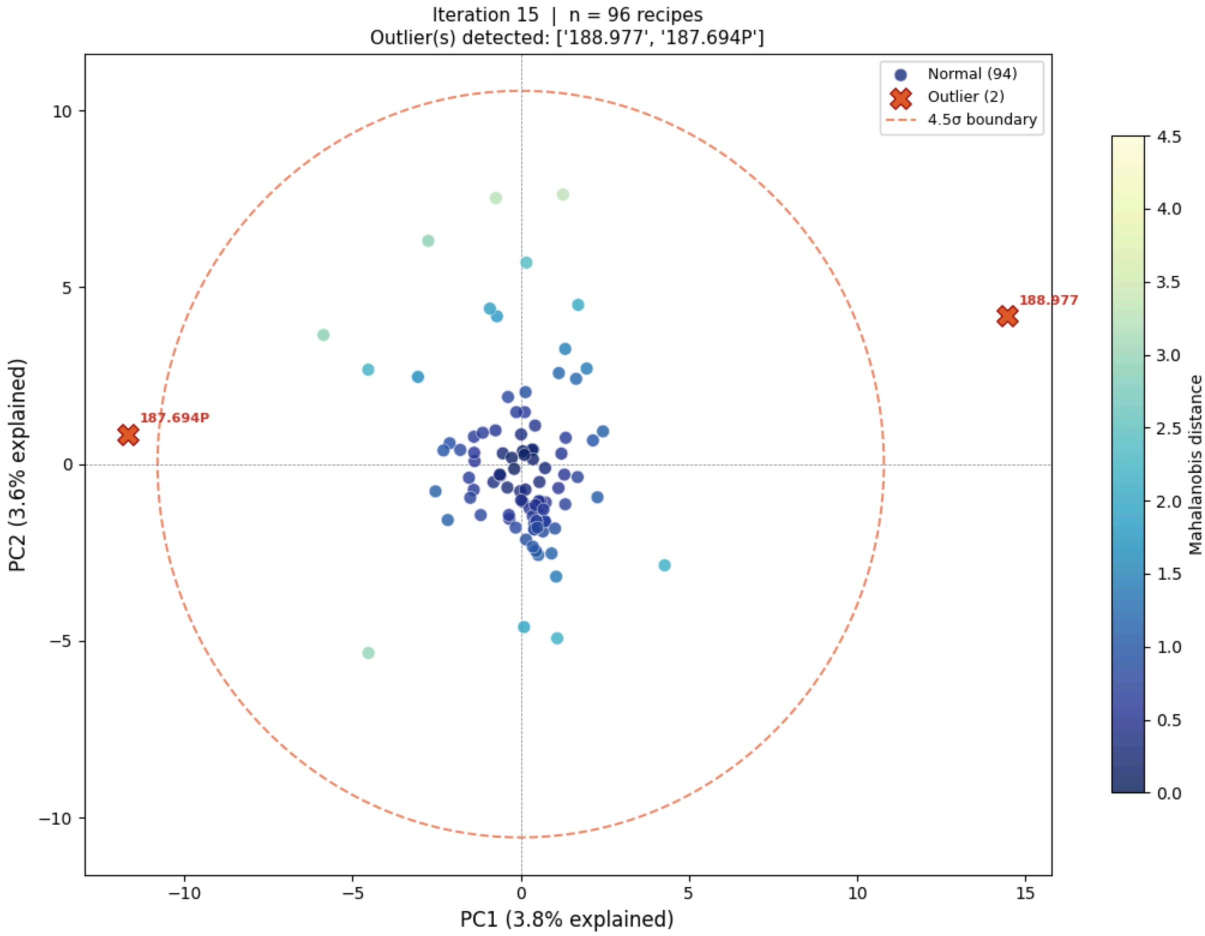

```
Saved: pca_v2_noOT_iter12.png
Iteration 12: removing 3 outlier(s): ['187.772', '187.746P', '185.043P']
```



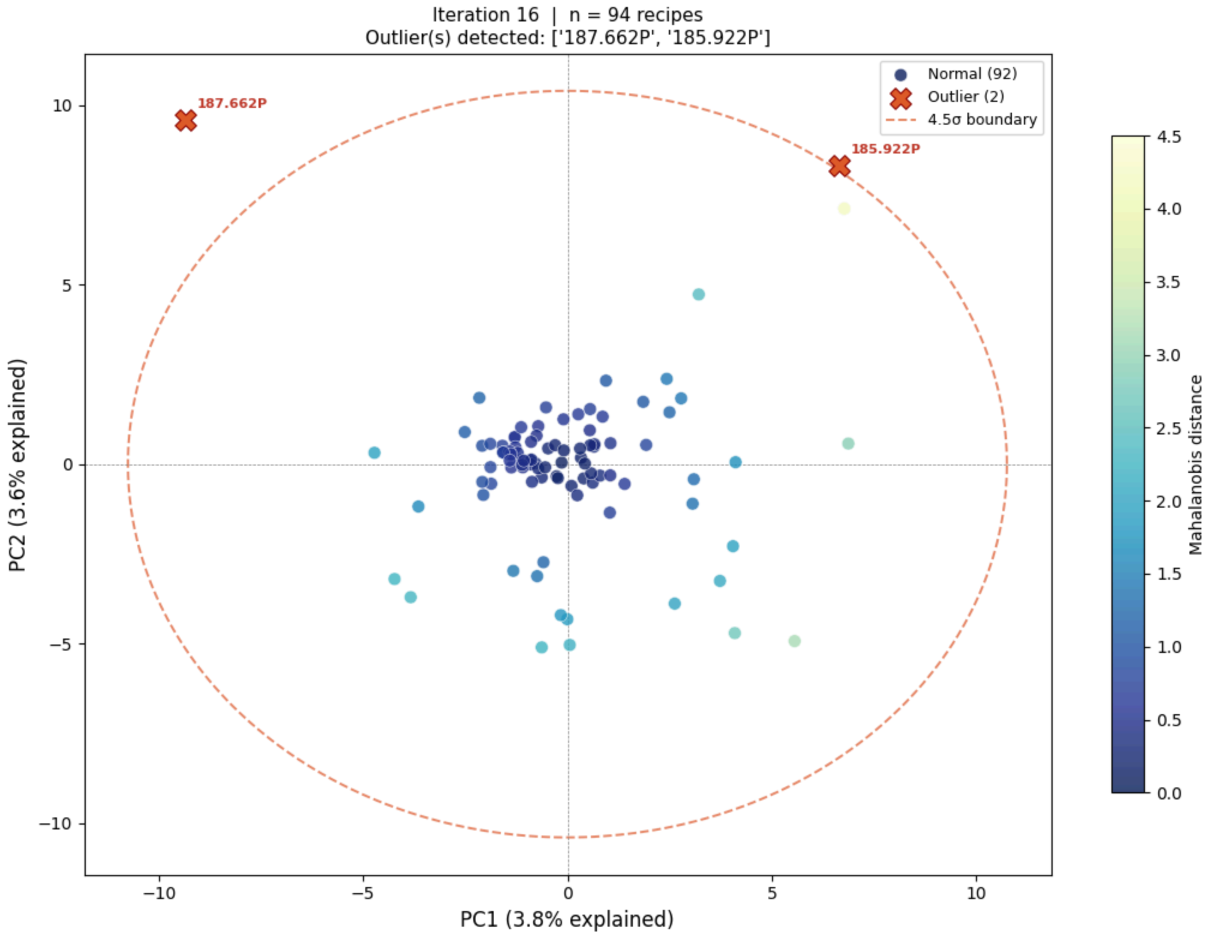

```
Saved: pca_v2_noOT_iter13.png
Iteration 13: removing 3 outlier(s): ['188.819', '185.044', '189.051']
```



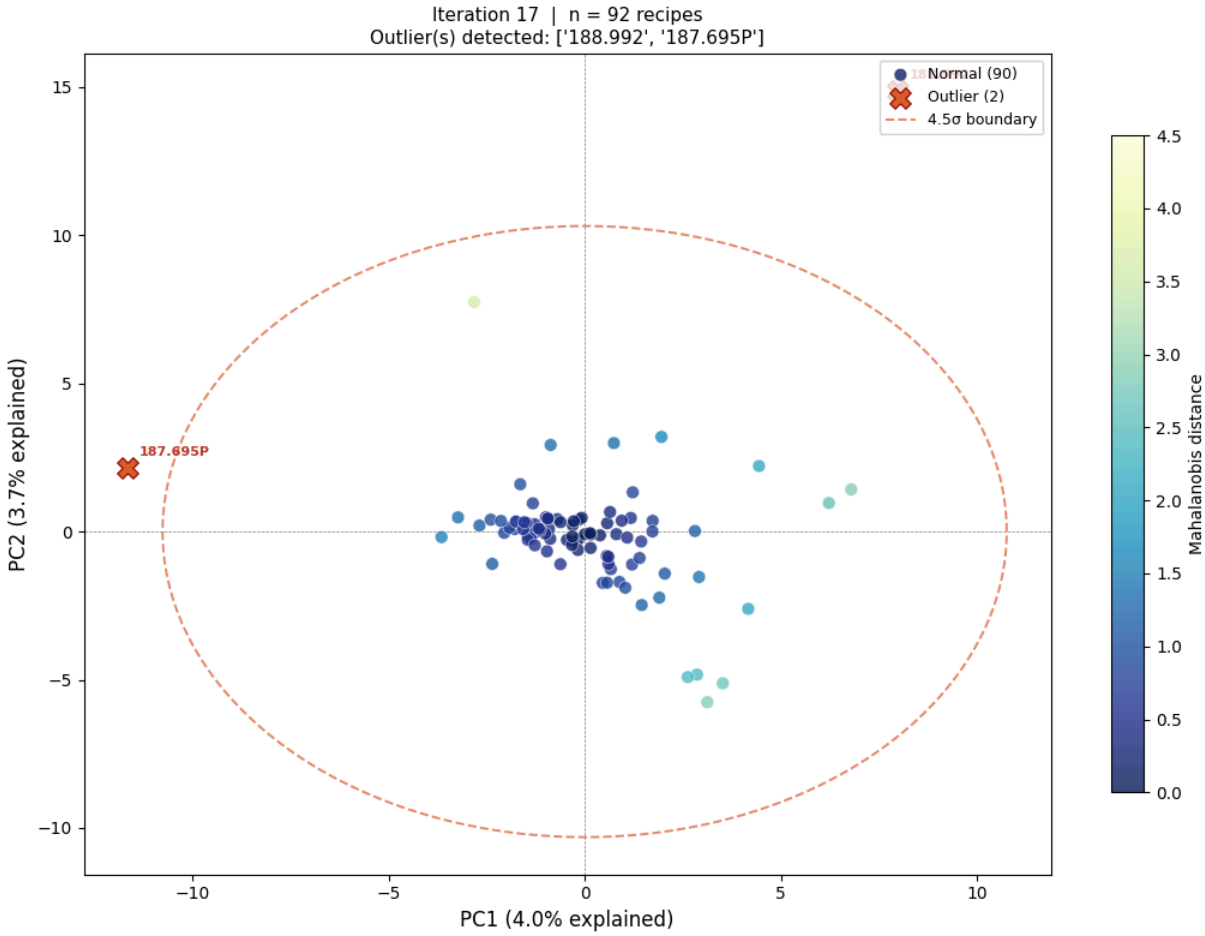

```
Saved: pca_v2_noOT_iter14.png
Iteration 14: removing 2 outlier(s): ['185.513P', '186.569P']
```



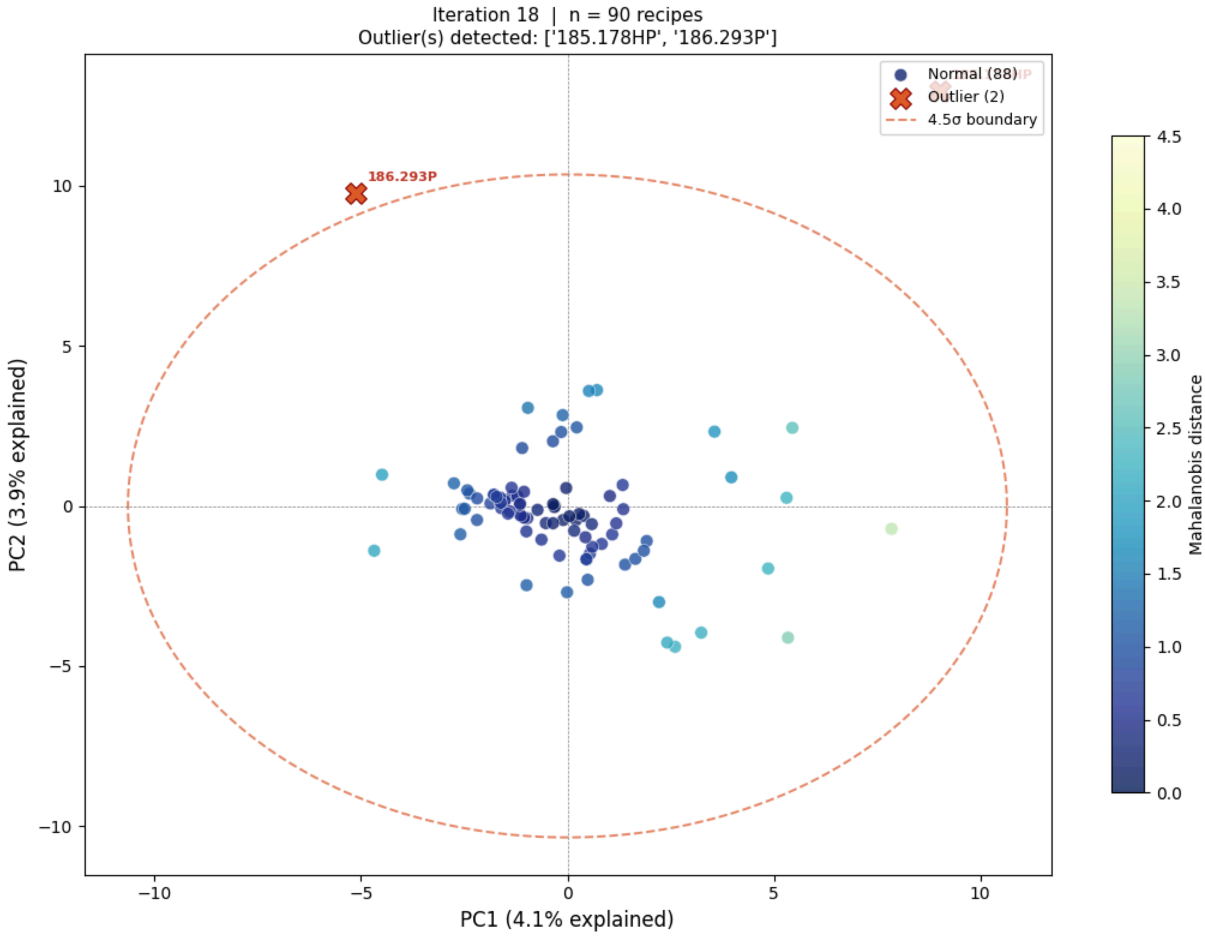

```
Saved: pca_v2_noOT_iter15.png
Iteration 15: removing 2 outlier(s): ['188.977', '187.694P']
```



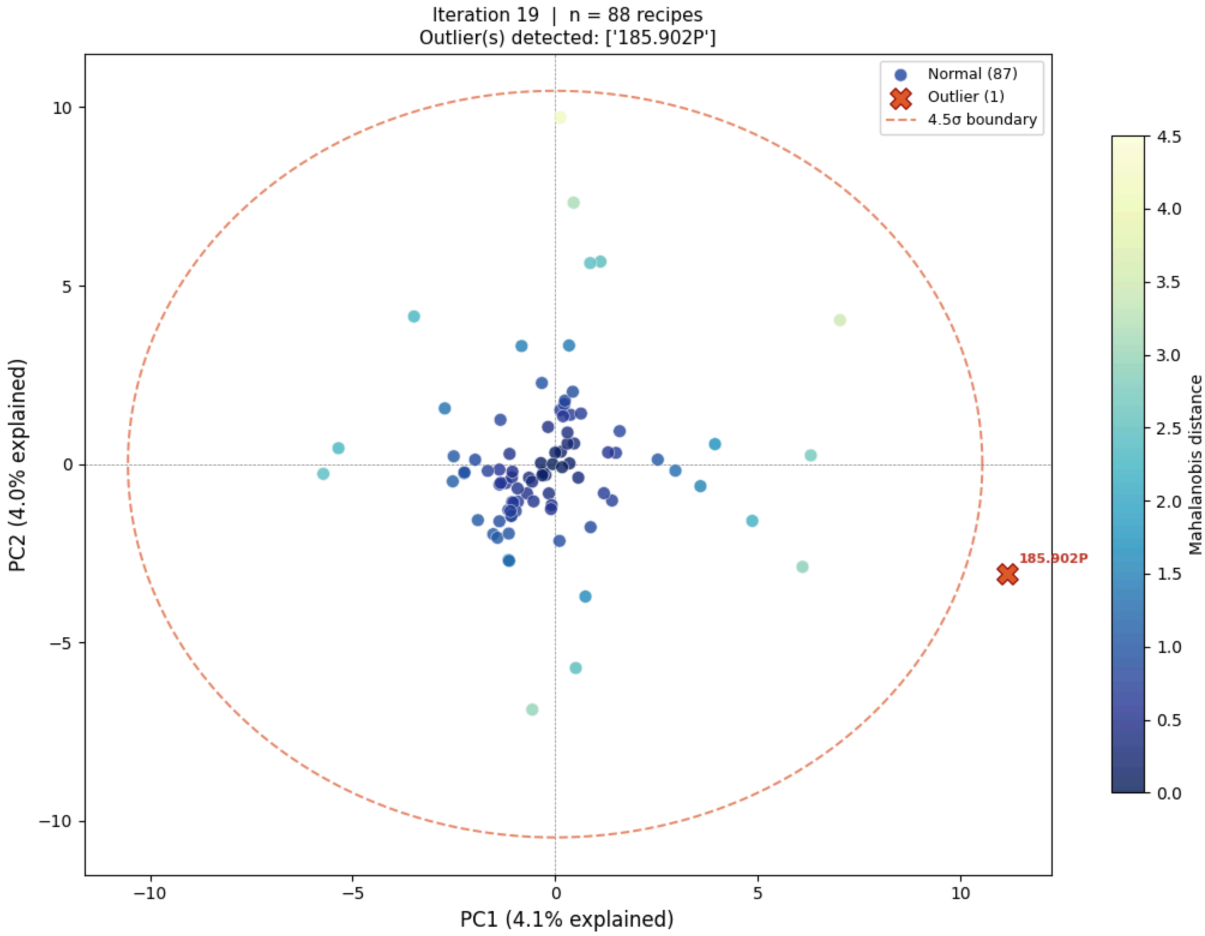

```
Saved: pca_v2_noOT_iter16.png
Iteration 16: removing 2 outlier(s): ['187.662P', '185.922P']
```



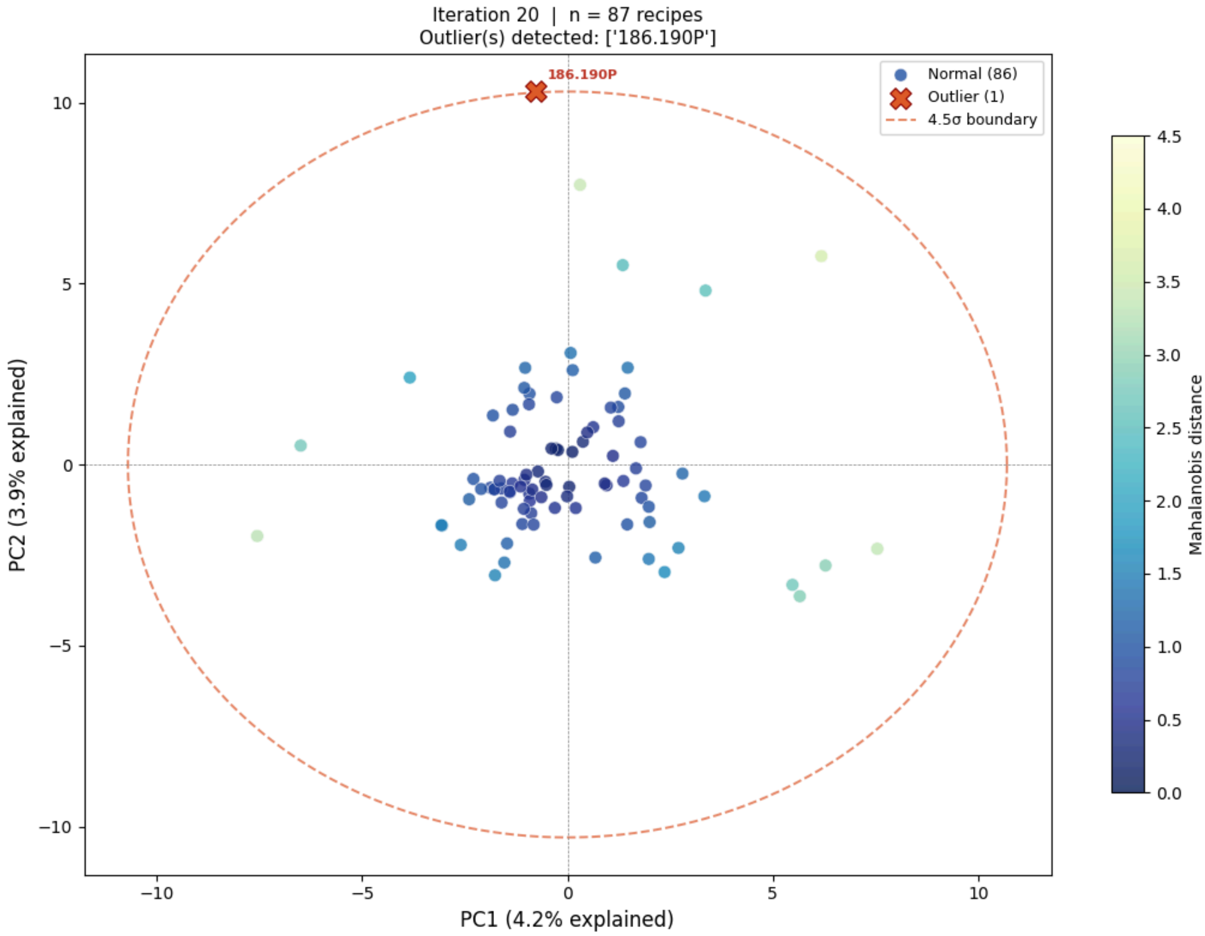

```
Saved: pca_v2_noOT_iter17.png
Iteration 17: removing 2 outlier(s): ['188.992', '187.695P']
```




```
Saved: pca_v2_noOT_iter18.png
Iteration 18: removing 2 outlier(s): ['185.178HP', '186.293P']
```




```
Saved: pca_v2_noOT_iter19.png
Iteration 19: removing 1 outlier(s): ['185.902P']
```




```
Saved: pca_v2_noOT_iter20.png
Iteration 20: removing 1 outlier(s): ['186.190P']

⚠ Maximum 20 iterations reached — stopping.
  Remaining dataset: 86 recipes

Total removed: 43 recipe(s)
Removed list: [(1, '187.004P'), (1, '185.309P'), (2, '187.015P'), (2, '187.4
82P'), (2, '185.382P'), (3, '188.462KHP'), (3, '185.028'), (4, '186.985P'),
(4, '185.748P'), (5, '186.950P'), (5, '186.490P'), (6, '186.179P'), (6, '18
6.395P'), (7, '188.628P'), (7, '187.507P'), (7, '186.396P'), (8, '187.894
P'), (8, '187.893P'), (8, '187.871P'), (9, '185.727P'), (9, '187.796P'), (1
0, '185.339P'), (10, '185.507P'), (10, '185.796P'), (11, '187.061P'), (12,
'187.772'), (12, '187.746P'), (12, '185.043P'), (13, '188.819'), (13, '185.0
44'), (13, '189.051'), (14, '185.513P'), (14, '186.569P'), (15, '188.977'),
(15, '187.694P'), (16, '187.662P'), (16, '185.922P'), (17, '188.992'), (17,
'187.695P'), (18, '185.178HP'), (18, '186.293P'), (19, '185.902P'), (20, '18
6.190P')]
```

In [8]:

```python
# ── Iteration summary table ──────────────────────────────────────
iter_df = pd.DataFrame(iteration_log)
print('=== Iteration Log ===')
print(iter_df.to_string(index=False))

# Set up convenient aliases used by all subsequent sections
scaler_oav   = final_scaler
pca_oav      = final_pca
scores_oav   = final_scores
```